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Catalytic behaviour of cobalt or ruthenium supported molybdenum carbide catalysts for FT reaction

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Abstract

One weight percent Co or Ru dispersed over Mo_2C have been investigated as catalysts for Fischer–Tropsch reaction and compared to bulk Mo_2C. It was found that Mo_2C gives light hydrocarbons, alcohols and CO_2. Noteworthy activity increases following the sequence: $Mo_2C < Ru/Mo_2C < Co/Mo_2C$. Compared with Mo_2C selectivity, the addition of Ru decreases the alcohol production whereas Co increases formation of heavy hydrocarbons. The carbon number distributions of hydrocarbons are consistent with a Schulz–Flory equation excepted for the cobalt based catalyst.

Keywords: CO hydrogenation; Molybdenum carbide; Cobalt; Ruthenium

1. Introduction

The production of fuels from natural gas is one of the most industrial challenge for near future. At present the process of methane conversion into valuable products occurs in two steps. Namely the first step is the conversion of methane to synthesis gas (H_2, CO). The second step is the conversion of syngas into valuable products such as heavy hydrocarbons and alcohols. In the first step, nickel catalysts are used industrially for both the methane stream and dry reforming reactions because of their fast turnover rates, long-term stability and low cost [1]. As for FT reaction, reaction proceeds on transition metal supported catalysts. The catalysts currently investigated being Co, Fe or Ru supported on oxides as Al_2O_3, SiO_2, TiO_2, ZrO_2 modified by SiO_2 and TiO_2 [2–7]. This shows that different catalysts must be used presently for the overall process. Direct conversion of methane into valuable products could be a new field of catalysis research, namely in the design of a new catalyst able to perform both steps of the process. For purpose, Green and coworkers have reported a few years ago that Mo_2C and WC could be potential alternatives to nickel catalyst in industrial process [8–10]. Investigation of Group VI transition metal carbides have also been carried out in FT reaction. Synthesis of hydrocarbons from CO and H_2 has been evidenced by Kojima et al. [11] on metal carbide catalysts. The rise in CO hydrogenation activity among molybdenum carbides and nitrites has been interpreted by Ranhotra to a progressive loss of oxygen from the catalyst surface [12]. Leclercq et al. [13] and Woo et al. [14] have elsewhere clearly shown that tungsten and molybdenum carbides produced mainly light alkanes, whereas the formation of alcohols is related to the surface stoichiometry and to the extent of carburization [13]. Promotion of Mo_2C with K_2CO_3 has been found to greatly enhance the selectivity to alcohols composed of linear C1–C7 [14]. It thus appears that molybdenum and tungsten carbides could be potential catalysts for the direct methane transformation into valuable products such as heavy hydrocarbons. This requires among others things an optimization of the carbide catalysts for the FT reaction. To achieve this goal, one possibility could the addition of small amounts of a Fischer–Tropsch synthesis metal such as Co or Ru on Mo_2C.

Thus, the aim of this work is to study the influence of the addition of Co or Ru at low loading (1 wt.%) on Mo_2C in terms of activity and selectivity in FT reaction compared with those of the parent molybdenum carbide. The catalysts have been characterized by elemental analysis, nitrogen adsorption, X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS) and tested in the CO–H_2 reaction.

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2. Experimental

2.1. Preparation of the catalysts

2.1.1. Synthesis of molybdenum carbide

Molybdenum carbide was prepared by Temperature Programmed Experiment. MoO_3 (10 g) (Merck, 99.5% purity) was reduced and carburized in a mixture of 10% CH_4–H_2 at a flow rate of $10\,l\,h^{-1}$ from room temperature to a final value of 973 K ($\beta = 60\,K\,h^{-1}$). The final temperature was maintained for about 6 h. After carburization, the sample was submitted to a flow of N_2 ($3\,l\,h^{-1}$) for 30 min and finally passivated at room temperature in a 2% O_2–He mixture for 2 h ($2\,l\,h^{-1}$) to protect the bulk of the catalyst against deep oxidation. The procedure was carried out twice. Although a small amount of Mo metal was detected by XRD in the first preparation, both powders exhibit on the other hand rather similar nominal composition as determined by XPS measurement (cf. 3.1.1). As XPS surface states reproducibility has been achieved for both solids, the samples were labeled Mo_2C.

2.1.2. Synthesis of supported catalysts

The catalysts were obtained by wet impregnation of the molybdenum carbide with an aqueous solution of respectively ruthenium chloride dihydrate (Fluka, purum, ≈38–40% Ru) or cobaltous nitrate hexahydrate (Fluka, ≥98% purity), to obtain a nominal content of 1 wt.% Ru or Co. After slow evaporation of the solvent, the solids were dried at 393 K overnight. Hereafter the solids will be denoted Y/Mo_2C, where Y is the Co or Ru metal.

2.2. Physical characterizations

2.2.1. Elemental analysis

The chemical analysis of the supported catalysts were determined by atomic absorption for Co, Mo and Ru and by coulometry for C, by the Central Service of Chemical Analysis of the CNRS (Vernaison, France).

2.2.2. Surface area

The BET surface areas were measured by a single point BET method using a QUANTASORB J.R. apparatus. Before experiment, the fresh catalysts were outgassed in a flow of nitrogen for 30 mn at 423 K, in opposition to the tested solids, which were outgassed at 533 K for 1 h.

2.2.3. X-ray diffraction

XRD patterns were recorded at room temperature by a SIEMENS D5000 diffractometer using Cu Kα radiation.

2.2.4. XPS

The X-ray photoelectron spectra (XPS) were recorded with a VG ESCALAB 220XL spectrometer equipped with a monochromatized Al source (Al Kα = 1486.6 eV). The analyser was operating in a constant pass energy mode ($E_{pas} = 30$ eV) using the electromagnetic mode for the lens. The resolution measured on Ag $3d_{5/2}$ peak was 0.75 eV. Energy correction was performed using the C 1s peak at 285 eV as a reference. Spectra of all solids were recorded before and after reduction in flowing hydrogen. The reduction of the catalysts was carried out in a flow of pure hydrogen ($2\,l\,h^{-1}$) in the preparation chamber close to the analysis chamber, at the temperature adopted for the activation treatment before catalytic test (Mo_2C and Ru/Mo_2C: 673 K-12 h; Co/Mo_2C: 773 K-5 h) ($\beta = 3\,K\,min^{-1}$).

2.3. Catalytic activity measurement

The catalytic tests were performed in a stainless steel fixed-bed flow reactor [5,15] operating at 473–533 K and total pressure of 20 or 50 bar with VSV ranging from 3150–7400 h^{-1} (VSV: volumetric space velocity defined as the reactant gas flow divided by the catalyst volume). Hydrogen (99.995%, Air liquide) and carbon monoxide (99.94%, Air liquide) were supplied to the reactor through mass flow controllers (Brooks). The H_2/CO ratio was 2 in all experiments. N_2 ($0.3\,l\,h^{-1}$) was used as internal standard. The catalysts loadings were of about 2 g.

Prior to the catalytic test, all the samples were activated in a flow of pure hydrogen ($3.6\,l\,h^{-1}$, 12 h) at atmospheric pressure from room temperature either to 673 K for Mo_2C and Ru/Mo_2C or to 773 K for Co/Mo_2C. The catalysts were then cooled to the initial reaction temperature (473 K), and the pressure was increased in a H_2–N_2 flow up to 20 bar. Then the H_2 and N_2 flows were adjusted to the values of the reaction test and CO was introduced. To avoid a possible condensation of the reaction products, gas transfer lines were continuously heated at 393 K. Analysis of the gaseous products were carried out on line with a gas chromatograph (Varian 3400) equipped with TCD and FID detectors with CTR-1 for C1 products and a Tenax column for hydrocarbons (C_1–C_{10}) and alcohols up to C_3, respectively. High-molecular-weight products (C_{10}^+ hydrocarbons) were collected from a hot condenser heated at 393 K. The wax analysis was performed on a WCOT ULTI-METAL column (coating HT SIMDIST CB). Catalytic rates and selectivities were measured at the stationary regime after circa 20 h time-on-stream. The conversion *X*, expressed in percentage, is the ratio of the number of moles of CO converted to the initial number of moles of CO. Specific reaction rates, expressed in mol $h^{-1}\,g^{-1}$, were defined as the number of moles of CO converted per unit time per gram of catalyst. Product selectivity (*S*) was reported as the percentage of CO converted into a given product expressed in C atoms, excluding CO_2. $S(C_n)$ and $S(C_5^+)$ were referred, respectively, to the selectivity in hydrocarbons with n carbon atoms and to the selectivities of all hydrocarbons in the gas phase with a carbon atom number higher than or equal to 5. By the same way, *S*(COH) is the global selectivity in alcohols. The proportion (%) of ethanol and propanol in alcohols is denoted C_2OH^+. Carbon mass balanced are respected within the margin of

Table 1
Characterization of the catalysts

Catalyst	Me (%)	C (%)	Mo (%)	Atomic ratio C/Mo	S_{BET} ($m^2\,g^{-1}$)
Mo_2C	–	3.6([a])–5.5	90.8([a])–88.5	0.3([a])–0.5	9.2([a])–6.7
Ru/Mo_2C	0.78	4.5	91.5	0.4	7.3
Co/Mo_2C	0.97	5.2	86.0	0.5	3
Mo_2C tested	–	17.4	74.2	1.9	2.9
Ru/Mo_2C tested	0.87	21.4	71.8	2.4	4.3
Co/Mo_2C tested	0.70	26.5	68.5	3.1	2

[a] First preparation.

error of around 20% for all catalysts. Ln(S/n) was plotted versus the carbon number n, in order to see if the products composition followed a Schulz–Flory distribution [16].

3. Results

3.1. Catalyst characterizations

3.1.1. Fresh catalysts

3.1.1.1. Mo_2C. Chemical analysis and surface areas results are reported in Table 1. The X-ray diffraction patterns (not presented here) were that of Mo_2C with a hexagonal close-packed structure. Noticeable a small contribution of Mo metal was observed on the pattern of Mo_2C originating from the first preparation. This is in agreement with excess of molybdenum observed by chemical analysis (C/Mo = 0.3), on contrary to the second one having the expected stoichiometry (C/Mo = 0.5). The total surface area of the catalysts was low ($<10\,m^2\,g^{-1}$), but the aim of the preparation was not to optimise the specific surface of the material.

Surface characterization has been performed by XPS (Table 2). For both preparations, the Mo 3d spectrum (Fig. 1a) exhibits a complex envelop in accordance with different Mo components at the surface. The peak Mo $3d_{5/2}$ at the lower binding energy (228.2 eV) is consistent with the carbidic phase [17], no metallic molybdenum being

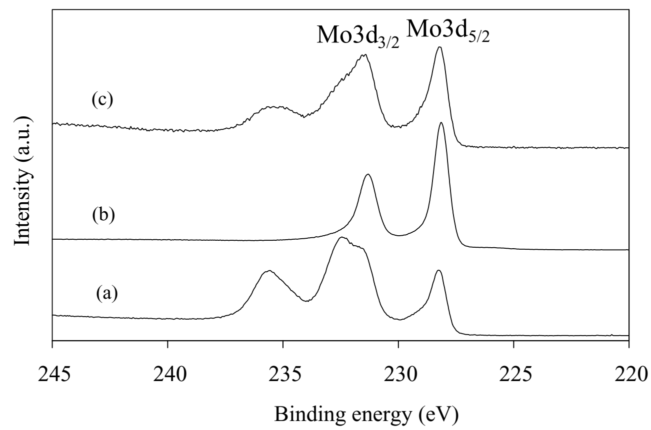

Fig. 1. Mo 3d XPS spectra of Mo_2C. (a) Before reduction in H_2, (b) after reduction in H_2 and (c) used catalyst.

detected. The remaining peaks are characteristic of oxidized phases, resulting from the passivation step. The C 1s spectrum shows three peaks which evidence carbidic (B.E. = 283.3 eV [17]), free (B.E. = 285 eV) and oxidized carbons (B.E. at about 289 eV) (Fig. 2a).

As XPS surface states of the two materials are rather identical, it has been concluded that molybdenum metal detected in the first preparation is rather in the core of the particle. So for catalytic purpose, the two solids have been considered as to be similar.

Table 2
XPS results for the catalysts before and after reduction in H_2

Compound	Element	B.E. (eV) before reduction	B.E. (eV) after reduction
Mo_2C	$Mo3d_{3/2}$	235.6	231.3
	$Mo3d_{5/2}$	228.2	228.1
	C1s	283.3–285–289	283.4–285
Ru/Mo_2C	$Mo3d_{3/2}$	235.6	231.3
	$Mo3d_{5/2}$	228.2	228.1
	$Ru3d_{5/2}$	281.9	280
Co/Mo_2C	$Co2p_{3/2}$	781.0	778.2
	$Co2p_{1/2}$	796.6	793.2
	$Mo3d_{3/2}$	235.7	231.3
	$Mo3d_{5/2}$	228.2	228.1
	C1s	283.3–285–289	283.3

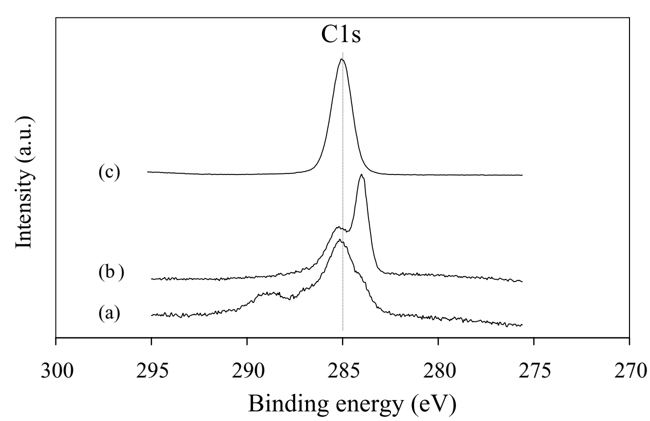

Fig. 2. C 1s XPS spectra of Mo_2C. (a) Before reduction in H_2, (b) after reduction in H_2 and (c) used catalyst.

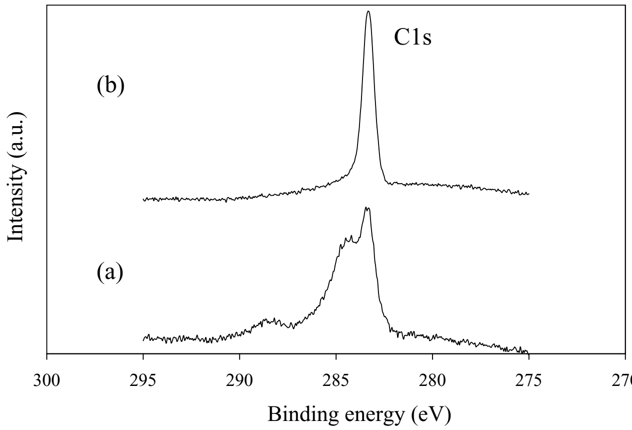

Fig. 3. C 1s XPS spectra of Co/Mo_2C. (a) Before reduction in H_2 and (b) after reduction in H_2.

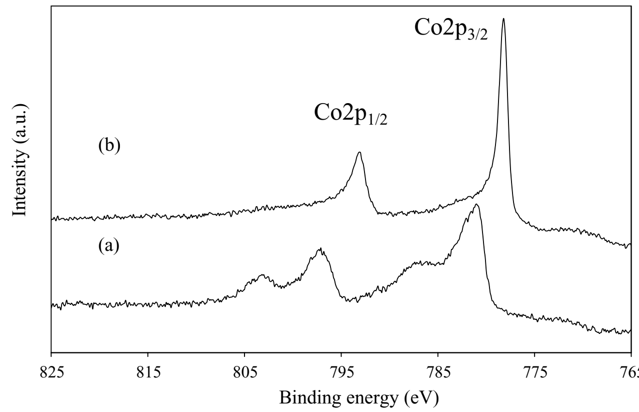

Fig. 5. Co 2p XPS spectra of Co/Mo_2C. (a) Before reduction in H_2 and (b) after reduction in H_2.

3.1.1.2. Co and Ru on Mo_2C. The X-ray diffraction patterns of the as prepared impregnated samples show only the Mo_2C with a hexagonal close-packed structure. No related metal (Co or Ru) phases were detected which is not surprising taking into account the low metal content. Co weight percentage was almost equal to the expected 1 wt.% value, Ru content being slightly lower due probably to the uncertainty of Ru content in the starting salt. Additionally, a decrease of the total surface area was observed for supported catalysts, due to the Co or Ru impregnation.

As expected, by XPS, both the C1s (Figs. 3a and 4a compared to Fig. 1a) and Mo 3d (Table 2) envelops of the impregnating samples are consistent with the ones of Mo_2C. The ruthenium exhibits a Ru $3d_{5/2}$ peak at 281.9 eV, characteristic of Ru^{3+} [18] (Fig. 4a). Chlorine is also detected. The Cl 2p signal shows a peak at 198.5 eV having a shoulder at lower binding energy 199.7 eV, with a XPS Cl/Ru ratio equal to 2.8. The Co $2p_{1/2}$ and Co $2p_{3/2}$ binding energies, respectively, at 796.6 and 781.0 eV, and the intense shake-up satellite structures are in agreement with the presence of Co^{2+} ions at the surface (Fig. 5a) [19,20].

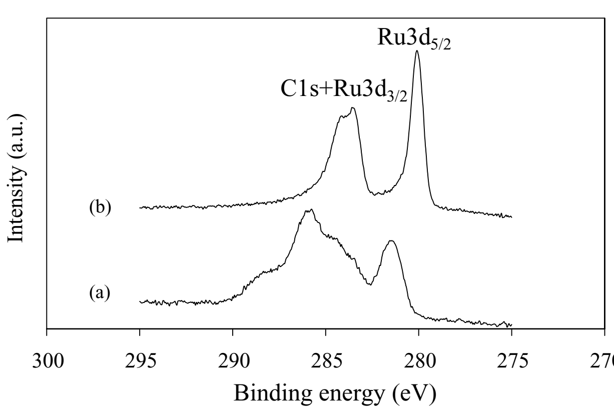

Fig. 4. C 1s and Ru 3d XPS spectra of Ru/Mo_2C. (a) Before reduction in H_2, (b) After reduction in H_2.

3.1.2. Catalysts activated by hydrogen

For Mo_2C, activation by pure H_2 at 673 K for 12 h allows to get rid of the passivation layer before reaction [21]. The Mo 3d signal (Fig. 1b) exhibits only two photopeaks separated by 3.2 eV indicating the total removal of the oxidic molybdenum components. Noteworthy is the disappearance of adsorbed carbonaceous species.

The reduction temperature of the supported samples has been chosen from ATG measurements based results performed in pure flowing H_2, which have shown that $RuCl_3$, xH_2O is totally reduced at 623 K on contrary to Co_3O_4 which is partially reduced into CoO and Co(0). Thus, the reduction temperature has been chosen at 673 K for Ru/Mo_2C and 773 K for Co/Mo_2C in order to ensure to get the metal species totally reduced at the lowest temperature. To validate the efficiency of the activation treatment, that is the total conversion of Ru and Co oxide species into metal, reduction in flowing H_2 (cf. 2.2) has been monitored by XPS. The Ru $3d_{5/2}$ peak shifts to lower B.E. value that is 280 eV, characteristic of Ru(0) [18,22,23] (Fig. 4b). Decomposition treatment allows to recover a carbidic component at 283.4 eV, free of carbon of pollution. On the Co 2p spectrum (Fig. 5b) a well resolved doublet is observed. A cobalt metal phase is attested by the B.E. at 778.2 eV Co $2p_{3/2}$ photopeak [20,24–26]. The C 1s signal (Fig. 3b) shows only one peak at 283.3 eV, corresponding to carbidic phase. For the two reduced supported catalysts, the Mo 3d signal (Fig. 1b) exhibits only two photopeaks separated by 3.2 eV. The value of the Mo $3d_{5/2}$ level at B.E. of 228.2 eV readily identify the carbidic phase. It is noticeable that for Co/Mo_2C the quantitative analysis leads to a surface molybdenum carbide stoechiometry C/Mo of 0.45.

Thus, XPS data show that after activation on the supported catalysts, Co and Ru are totally reduced on a clean Mo_2C surface.

3.1.3. Used catalysts

For all catalysts XRD patterns of the tested catalysts show that the structure of molybdenum carbide is preserved after

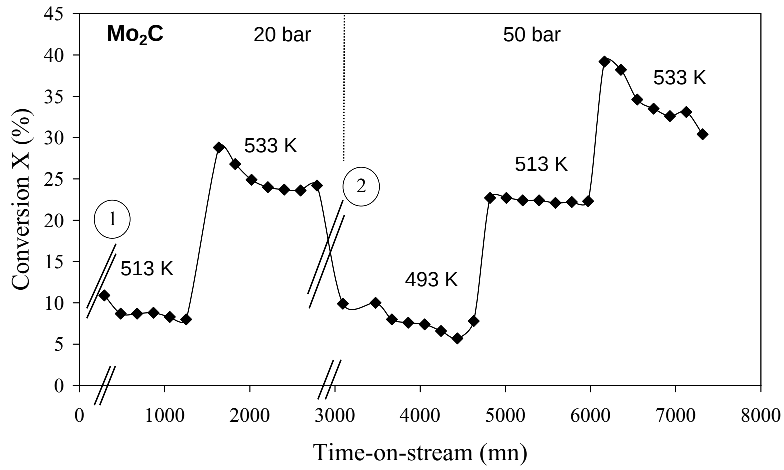

Fig. 6. Time-on-stream behaviour of conversion over Mo_2C catalyst. (①): Catalytic test is performed at 20 bar, respectively, at 473 and 493 K for 24 h. ② : Reaction was stopped. Catalyst remains in flowing H_2 for 48 h at atmospheric pressure and 473 K. Then catalytic test is performed at 473 K and 50 bar for 24 h.

tests. Further, we observe a substantial decrease of the surface area to reach a mean value of 2–3 $m^2\,g^{-1}$. This trend could be linked to the presence of an excess of carbon on the catalysts. This is confirmed by elemental analysis results (Table 1). XPS has also been performed on the used Mo_2C sample. The C 1s signal shows an intense peak of free carbon which hides the carbidic component (Fig. 2c). A direct comparison of the ratios $(C/Mo)_{XPS}$ of 33 and $(C/Mo)_{EA}$ 1.9 shows that carbon deposit occurs at the surface of the material. The Mo 3d envelop is relatively similar to the one of the fresh catalyst (Fig. 1c). It shows nevertheless minor oxidic components as free carbon protects the carbidic phase from deep reoxydation.

3.2. Catalytic behaviour

3.2.1. Study of Mo_2C

3.2.1.1. Influence of the temperature of reaction. CO conversion with time-on-stream was determined allowing temperature (473–533 K by step of 20 K) and pressure (20 or 50 bar) to vary ($H_2/CO = 2$) (Fig. 6). The experiment has been proceeded in stages and allows to vary one parameter, i.e. temperature or pressure. For each temperature stage, one chromatogram is obtained every 3 h. Conversion and selectivities reported in Table 3 have been obtained at steady state, when relative variation of conversion is less than 5% and the carbon selectivities unchanged within the margin of error, that is for a duration of around 20 h. At 20 bar for the two lower temperatures, the conversion is very low (≤1%, cf. Table 3) and as been omitted on the Fig. 6 for sake of clarity. A temperature increase from 473 to 533 K induces an increase in CO hydrogenation activity, the apparent activation energy being of 170 $kJ\,mol^{-1}$.

Product distribution on Mo_2C at 473 K and 20 bar (Table 3) is as follow: mainly light alkanes (C_1–C_4: 71%), higher alkanes up to C10 (C_5^+: 10%) and alcohols (COH: 19%). Ln(S/n) versus the carbon number n plots show a linear behaviour (Fig. 7). As the products distribution agrees with the Schulz–Flory model, the reaction of formation of hydrocarbons corresponds to a polymerisation of $=CH_x$ species. As temperature increases from 493 to 533 K, production of light alkanes increases, as attested by a decrease of α from 0.41 to 0.33. Besides, a decrease of alcohols

Table 3
Catalytic behaviour of Mo_2C catalyst ($m_{cata.} = 2.0406$ g)

T (K)	P (bar)	VSV (h^{-1})	X (%)	Rate (10^{-3} $mol\,h^{-1}\,g^{-1}$)	$S(CO_2)$ (%)	$S(C_1)$ (%)	$S(C_2–C_4)$ (%)	$S(C_5^+)$ (%)	S(COH) (%)	C_2OH^+ (%)	α
473	20	3150	<1	<0.4	∼0	32.1	38.6	9.9	19.4	44	–
493	20	3150	1.1	0.43	28.8	30.1	44.4	8.6	16.9	25	0.41
513	20	3150	8.1	3.17	31.4	37.9	50.6	7.1	4.4	10	0.38
533	20	3150	24.2	9.45	32.6	54.6	41.4	4.0	0	0	0.33
473	50	3150	2.1	0.82	11.7	31.3	34.9	7.6	26.2	41	0.43
493	50	3150	7.8	3.06	16.4	34.6	44.2	10.4	10.8	42	0.42
513	50	3150	22.3	8.74	21.8	41.4	48.1	7.0	3.5	23	0.39
533	50	3150	33.1	12.97	16.6	57.8	33.6	7.8	0.8	14	0.33
493	50	5550	4.4	3.23	14.6	31.7	42.6	9.2	16.5	35	0.42
513	50	5550	8.9	6.54	21.2	39.3	47.8	7.4	5.5	22	0.39

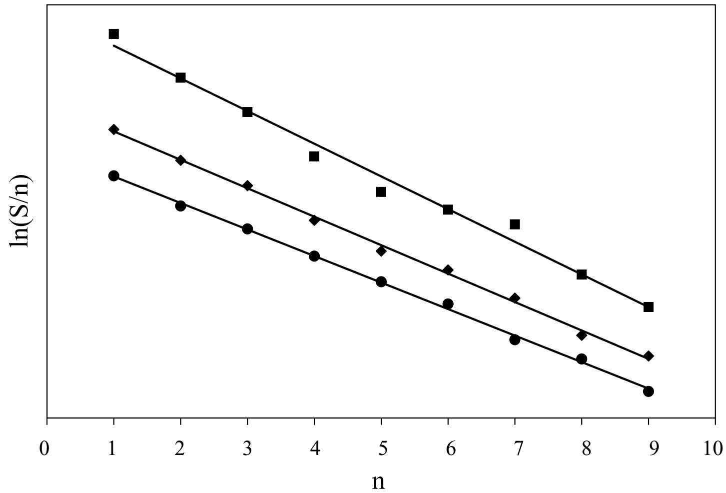

Fig. 7. Schulz–Flory plots of the Mo_2C catalyst distribution at 20 bar for different temperatures: (●) 493 K, (◆) 513 K, (■) 533 K.

selectivity at the expense of high alcohols is observed. The same observation keeps valid at 50 bar. Noteworthy the influence of residence time on conversion does not reveal diffusional limitations. All these observations are consistent with previous works which have shown that the conversion as well as the production of light products are favoured by an increase in temperature of reaction [27].

Whatever the experimental conditions, it is noticeable that the CO_2 production amounts to 10–30 %. Whatever temperature value, CO_2 selectivity keep rather unchanged.

3.2.1.2. Influence of the pressure. These results show that, for a given temperature, rising the pressure from 20 to 50 bar leads to an increase of activity, the apparent activation energy decreasing from 170 to 120 kJ mol^{-1}, whereas the selectivities at isoconversion (comparing results for a VSV of 3150 and 5550 h^{-1}) were not significantly affected by the change of the pressure. Direct comparison at 513 K shows that an increase of the pressure induces a one third decrease of the CO_2 selectivity (31.4–21.2%).

3.2.2. Study of Ru and Co supported catalysts

3.2.2.1. Ru/Mo_2C. Conversion at 20 bar and 473 K is low but significant (Table 4). A temperature increase from 473 to 513 K induces an increase in hydrogenation activity, the apparent activation energy being of 150 kJ mol^{-1}. Conversion with time-on-steam shows that Ru/Mo_2C exhibits a rather good stability excepted at 513K, 50 bar (Fig. 8). This can either be due to a longer time stabilization or to a rapid deactivation of the catalyst. Decreasing further the temperature to 493 K for a higher VSV of 7400 h^{-1} exhibits the expected conversion, which readily demonstrates that the catalyst stabilization requires a longer time. At 473 K and 20 bar products are predominantly alkanes (98%) (C1–C4: 85%; C5–C10: 13%) with a very small contribution of alcohols (2%). Products distribution follows the Schulz–Flory model. A decrease of α from 0.41 to 0.37 is observed as temperature varies from 473 to 513 K. So again, increasing the temperature favors production of light products. As for Mo_2C, an increase of temperature reduces alcohol selectivity at the benefit of methanol. At 493 K an increase of the pressure induces a one third decrease of the CO_2 selectivity (29.1–19.5%), counterbalanced by an increase of alcohol selectivity. Variations of temperature and pressure (Table 4) induce similar trends than Mo_2C in the variations of activity and products selectivities.

3.2.2.2. Co/Mo_2C. At 20 bar and 473 K, Co/Mo_2C has a CO hydrogenation conversion five time higher than

Table 4
Catalytic behaviour of Ru/Mo_2C catalyst ($m_{cata.} = 1.8545$ g)

T (K)	P (bar)	VSV (h^{-1})	X (%)	Rate (10^{-3} mol h^{-1} g^{-1})	$S(CO_2)$ (%)	$S(C_1)$ (%)	$S(C_2–C_4)$ (%)	$S(C_5^+)$ (%)	S(COH) (%)	C_2OH^+ (%)	α
473	20	4200	1	0.43	~0	37.2	48.3	13.2	1.3	100	0.41
493	20	4200	6.1	2.63	29.1	40.0	48.3	11.5	0.2	100	0.42
513	20	4200	20.0	8.63	30.4	46.7	45.4	7.9	0	0	0.37
473	50	4200	4.3	1.85	12.8	35.0	37.6	11.9	15.5	48	0.46
493	50	4200	18.4	7.94	19.2	38.8	44.5	11.8	4.9	39	0.42
513	50	4200	32.7	14.11	16.9	57.3	34.2	7.3	1.2	35	–
493	50	7400	7.9	6.39	19.5	38.1	45.6	10.1	6.2	35	0.42
513	50	7400	14.1	11.40	18.3	54.7	36.8	7.7	0.9	33	0.37
513	20	7400	7.0	5.66	27.8	52.1	33.4	14.4	0	0	0.29

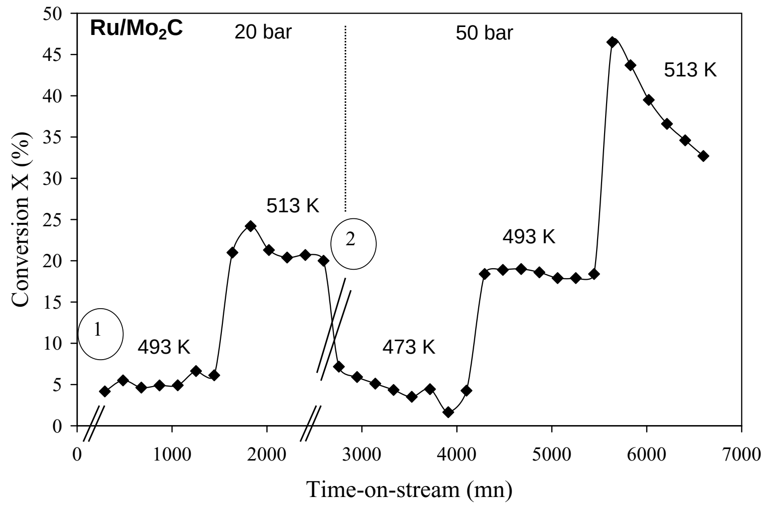

Fig. 8. Time-on-stream behaviour of conversion over Ru/Mo_2C catalyst. ➀ : Catalytic test is performed at 473 K and 20 bar for 24 h. ➁ : Reaction was stopped. Catalyst remains in flowing H_2 during 48 h at atmospheric pressure and 473 K.

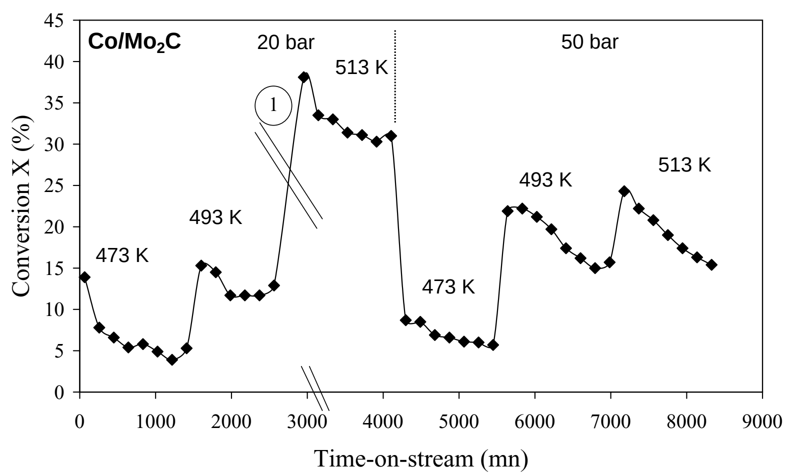

Fig. 9. Time-on-stream behaviour of conversion over Co/Mo_2C catalyst. ➀ : Reaction was stopped. Catalyst remains in flowing H_2 during 48 h at atmospheric pressure and 473 K.

Ru/Mo_2C. Noteworthy the apparent activation energy is the lowest value among the three catalysts (90 kJ mol^{-1}). Whereas the steady state is reached at 20 bar whatever the temperature step, at 50 bar the conversion decreases readily with time at a temperature of 493 and 513 K (Fig. 9, Table 5). A study of the effect of contact time on conversion shows a deactivation of the catalyst. At 473 K and 20 bar the gaseous phase product distribution on Co/Mo_2C is as

Table 5
Catalytic behaviour of Co/Mo_2C catalyst ($m_{cata.} = 2.0841$ g)

T (K)	P (bar)	VSV (h^{-1})	X (%)	Rate (10^{-3} mol h^{-1} g^{-1})	$S(CO_2)$ (%)	$S(C_1)$ (%)	$S(C_2–C_4)$ (%)	$S(C_5^+)$ (%)	S(COH) (%)	C_2OH^+ (%)	α
473	20	4200	5.3	2.03	25.5	30.6	41.4	15.2	12.8	39	–
493	20	4200	11.7	4.49	28.2	34.4	47.2	11.1	7.3	22	0.43
513	20	4200	31.0	11.90	29.4	50.4	44.0	5.6	0	0	0.35
473	50	4200	6.1	2.34	13.2	36.1	42.6	12.0	9.3	50	–
493	50	4200	15.7	6.03	17.5	42.9	43.1	9.6	4.4	36	0.37
513	50	4200	15.4	5.91	16.2	51.3	40.5	4.9	3.2	0	0.36
493	50	7400	5.8	4.17	16.8	44.6	41.3	8.0	6.1	34	0.39
513	50	7400	5.1	3.67	13.4	49.7	39.1	6.1	5.1	32	0.35
493	20	7400	3.7	2.66	23.9	40.1	35.1	14.4	10.4	19	0.41
513	20	7400	4.1	2.95	23.3	47.6	37.6	14.8	0	0	0.36

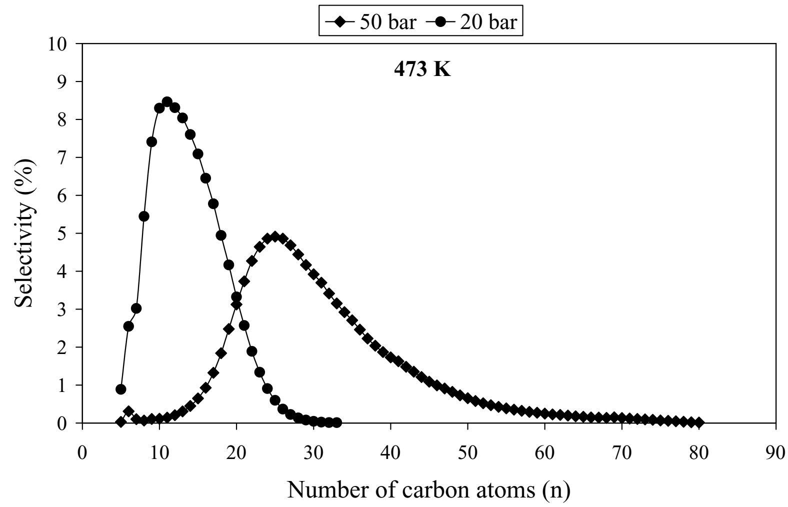

Fig. 10. Repartition of hydrocarbons products obtained in the hot condenser for Co/Mo_2C at 473 K.

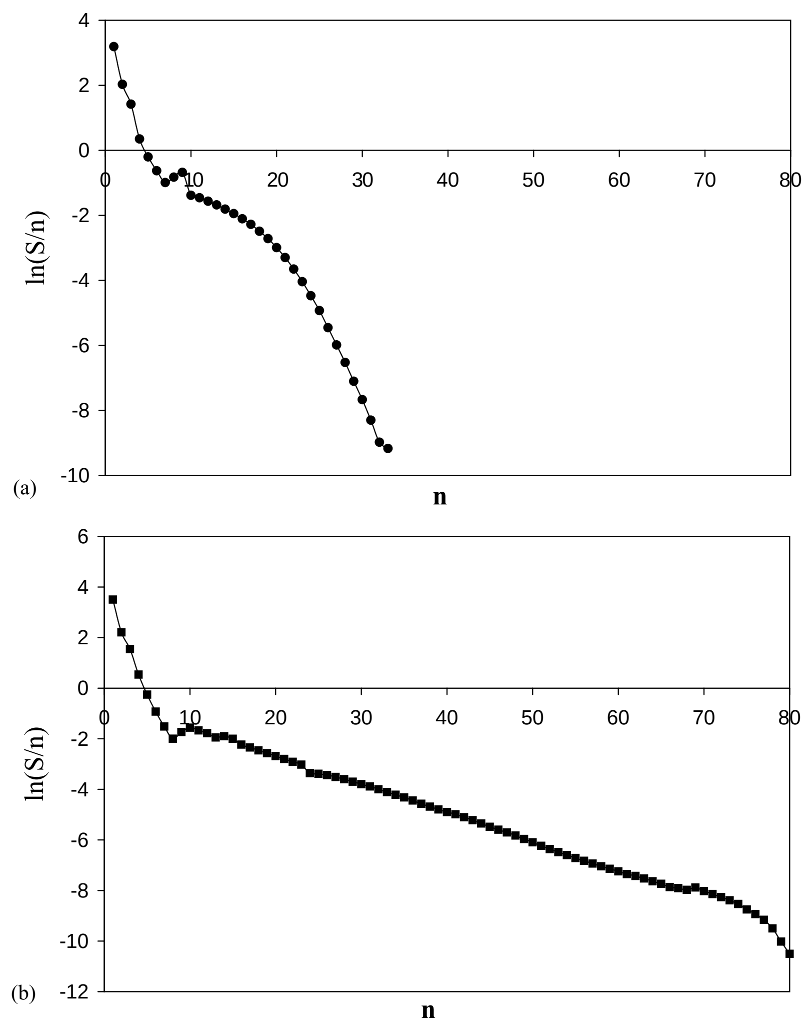

Fig. 11. Curve of $\ln(S/n)$ vs. n for Co/Mo_2C at 473 K: (a) 20 bar; (b) 50 bar.

follow: C1–C4: 72%, C5–C10: 15% and alcohols: 13%. The CO_2 selectivity is similar to those of Mo_2C and Ru/Mo_2C, either around 30% at 20 bar and 15% at 50 bar. Noteworthy here high hydrocarbons are trapped in the hot condenser (heated at 393 K) at 473 K whatever the pressure. The high hydrocarbons products distributions are given on Fig. 10. The hydrocarbons cuts at 20 and 50 bar are: C5–C33 and C5–C80, which are centered, respectively, at about 11 and 25 carbons atoms. Ln(S/n) versus n representations show complex plots whatever the pressure. At 20 bar linearity is observed up to C7 followed by a convex curve (Fig. 11a). At 50 bar the plot shows two segments of lines which intersect around C8; additionally a small break off is observed between C10–C20 (Fig. 11b).

4. Discussion

Group VI transition metal carbides could be attractive catalysts for the one step methane conversion into valuable hydrocarbon products, as molybdenum carbide is well-known to be active for the methane reforming and particularly for dry reforming to produce syngas. Few studies have been carried out on Mo_2C as Fischer–Tropsch catalyst [11,12,14]. It has been shown that molybdenum carbide promoted by K_2CO_3 can be a good catalyst for linear alcohol synthesis [14]. On the other hand, previous studies have shown that Mo_2C is a poor FT catalyst in the hydrogenation of CO into heavy hydrocarbons [11]. In order to convert in one step methane into heavy hydrocarbons, CO hydrogenation selectivity into $C_5{}^+$ is highly desirable. Thus, dispersed 1 wt.% of cobalt or ruthenium on Mo_2C materials have been prepared and tested in the FT reaction to be compared with Mo_2C. Indeed, these transition metals at metallic state are well-known to be highly selective into heavy hydrocarbons. Comparison of these materials in term of activity and selectivity has been carried out from Co or Ru species reduced at the lowest temperature that is 673 K for Ru and 773 K for Co.

Solids have been prepared by impregnation in excess from classical precursor that are $Co(NO_3)_2$, $6H_2O$ and $RuCl_3$, xH_2O. Activation of the catalysts under flowing H_2 has been performed by XPS in experimental condition as closed as the one performed before the catalytic test. Data show that Co and Ru are totally reduced on Mo_2C free of carbonaceous impurities. Based on chemical consideration, it is expected to have big islands of oxidized cobalt after impregnation considering both cobalt content and low S_{BET} value of the support. In order to have an idea about the dispersion of the metal after reduction, attempts have been made to determine the metal crystal size of the Co particle from XPS models. Extrapolating the different growth modes of fine metallic film on oxide substrates, redispersion can readily occur either as layer mode (Franck–van der Merwe) or a layer-island mode (Stanski–Krastanov) [28]. To validate one of these hypothesis the exponential attenuation model (Dreiling) [29] has been used. To model "the layer-island" mode, an addition parameter f has been introduced, which is the fraction of the support covered with cobalt particles, with the assumption that the distribution is homogeneous. The observed XPS intensity I_{Co}/I_S ratio of the cobalt dispersed phase, based on the Co 2p and Mo 3d areas, was used to estimate the particle size, d_{Co}, of Co according to the two simple procedures given in Appendix A. Calculation of the average size, d_{Co}, of the supported particles assumes that they are cubic with d_{Co} corresponding to the edge of the cubic particles. After simulation, irrelevant values have been obtained for the cobalt crystallite size, leading to reject the two XPS proposed models. Hence, it is strongly suggested that rather large particles of Co having an heterogeneity character are formed which can be easily reduced into Co metallic by pure H_2 at the experimental conditions.

Mo_2C shows activity for CO hydrogenation. Direct comparison with literature data are unfortunately impossible due to the different experimental conditions used in previous studies, that differ mainly by the pressure (1 bar [30], 80 bar [14]) and H_2/CO ratio values (3.7 [30], 0.98 [14]). Among the three catalysts, conversion increases as follow: $Mo_2C < Ru/Mo_2C < Co/Mo_2C$. At 20 bar and 473 K, the specific rates are, respectively: $<0.4 \times 10^{-3}$, 0.43×10^{-3} and 2.03×10^{-3} mol h^{-1} g^{-1}. Mo_2C favors production of light alkanes but also alcohols up to C3 and CO_2. Nevertheless the intrinsic selectivity of Mo_2C in CO/H_2 reactions is mainly towards hydrocarbons. In contrast, the formation of alcohols in the FT reaction can be strongly influenced by the extent of carburization. Indeed, after activation in flowing H_2 at 673 K prior to FT reaction, part of the surface oxygen probably reacts with carbidic carbon and is removed from the uppermost surface layer. At this lower temperature, carbon diffusion is not expected to occur. Hence, it is likely that the Mo_2C sample has some carbon vacancies which can both dissociate CO and allow CO insertion to form formyl or formate species. So, carbon vacancies can play a similar role as anion vacancies in oxides. Additionally, some strongly adsorbed oxygen and possibly subsurface oxygen present even after hydrogen pretreatment at 673 K [31] can be associated with promotion effect of reduce MoOx at the parameter of the carbide particle. This point of view has been already developped by Leclercq et al. for alcohol synthesis on WC samples [13]. Concerning CO_2 formation it is strongly suggested that water gas shift reaction can account for its production, as water has not been detected in substantial amount.

The introduction of Co and Ru on Mo_2C increases the conversion all the more the temperature increases, in agreement with the decrease of the apparent activation energy for CO hydrogenation (Mo_2C: 170 kJ mol^{-1} > Ru/Mo_2C: 150 kJ mol^{-1} > Co/Mo_2C: 90 kJ mol^{-1}). The effect of pressure is different owing to the catalyst. Conversion increases with pressure for Mo_2C and Ru/Mo_2C, whereas deactivation is observed on Co/Mo_2C, which can be explained taken into account the products distribution on this sample. Selectivity of Ru/Mo_2C is mainly toward hydrocarbon and CO_2 with a very small contribution of alcohol. Selectivity of Co/Mo_2C in gaseous phase is rather similar as a one of Mo_2C. Additionally, a liquid–solid phase characteristic of heavy hydrocarbons is observed. Noteworthy addition of Co increases notably the selectivity into heavy hydrocarbons, in contrast of Ru. Such a discrepancies was not expected as Co and Ru are well-known to dissociate CO on metallic site and to have a hydrogenating activity suitable to increase the chain size. Noteworthy as Ru/Mo_2C has been reduced as lower temperature than the Co/Mo_2C, it is expected to have a badly dispersion on the Co species. This is in line with XPS results which show that on Co/Mo_2C the cobalt is totally reduced at 773 K. Such a decrease of dispersion is believed to increase the selectivity towards heavy hydrocarbons [32]. It seems however difficult to explain such a discrepancy of selectivity behaviour only from simple consideration of dispersion change. It is speculated that Ru could interact with molybdenum as to enhance hydrogenation activity which would have as a result not to increase the chain length. The deactivation observed at 50 bar for Co/Mo_2C could be probably due to the saturation of catalyst pores by liquid reaction products. Indeed, an increase in total pressure at the same conversions and selectivities could result in condensation of hydrocarbons, which are normally in the gaseous state at atmospheric pressure.

5. Conclusion

One weight percent Co or Ru dispersed over Mo_2C have been investigated as catalysts for CO/H_2 reaction and compared to bulk Mo_2C. It was found that Mo_2C gives mainly light hydrocarbons, alcohols and CO_2. As Mo_2C intrinsic nature is to form light hydrocarbons, carbon vacancies or/and remaining oxygen adsorbed on the surface can account into alcohol formation. Additionally, Mo_2C is active for the water-gas shift reaction.

Addition of Ru or Co increases the activity following the sequence: $Mo_2C < Ru/Mo_2C < Co/Mo_2C$. Compared with Mo_2C selectivity, the addition of Ru decreases the alcohol production whereas Co increases formation of heavy hydrocarbons. The carbon number distributions of hydrocarbons are consistent with a Schulz–Flory equation excepted for the cobalt based catalyst.

Appendix A. Estimation of the CO particle size from XPS

The calculations were based on a simple model of cubic Co particles (edge, d_{Co}) deposited on a non porous support as described in [33]. The ratio of XPS intensity is

- Owing the layer attenuation model:

$$\frac{I_{Co}}{I_S} = \left[\frac{n_{Co}}{n_S}\right]\left[\frac{\sigma_{Co}}{\sigma_S}\right]\left[\frac{T(E_{kinet.,Co})}{T(E_{kinet.,S})}\right] \times \left[\frac{\lambda_{Co\to Co}}{\lambda_{S\to S}}\right]\left[\frac{1-\exp(-d_{Co}/\lambda_{Co\to Co})}{\exp(-d_{Co}/\lambda_{S\to Co})}\right]$$

- Owing the layer attenuation model considering a geometric coverage on the support f:

$$\frac{I_{Co}}{I_S} = \left[\frac{n_{Co}}{n_S}\right]\left[\frac{\sigma_{Co}}{\sigma_S}\right]\left[\frac{T(E_{kinet.,Co})}{T(E_{kinet.,S})}\right]\left[\frac{\lambda_{Co\to Co}}{\lambda_{S\to S}}\right] \times \left[\frac{1-\exp(-d_{Co}/\lambda_{Co\to Co})}{((1-f)/f)+\exp(-d_{Co}/\lambda_{S\to Co})}\right]$$

where f is the fraction of the support covered with Co. f can be calculated by the formula: $f = x/((1-x)\times\rho\times S\times d)$, x is the Co weight on 1 g catalyst, ρ the Co density and S the support specific surface area.

nCo, and nS are the densities of Co atoms and of the Mo of Mo_2C,σ are the cross-sections, T is the transmission factor of the spectrometer (T is proportional to the kinetic energy E_{kinet}. of the photoelectrons corresponding to the levels considered), $\lambda_{X\to Y}$ is the mean free path of the electrons emitted to the element X in a matrix Y, and d_{Co} is the particle size of Co. The levels considered were Co 2p and Mo 3d. The values of the cross-sections σ were found in ref. [34]. The levels considered were Co 2p and Mo 3d. Those of $\lambda_{X\to Y}$ were estimated considering the software MFP NIST [35].

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